

A Students who has just entered the portals of colleges finds it difficult to understand the subject taught to him/her this difficulty is mainly due to understand standard references. While preparing this "Allied practical manual Non- chordate, chordate and General principles of zoology" has been prepared to serve the special purpose of college students. This Allied practical Manual explains the basic Non- chordates, Chordates and General principles of Zoology in a simplified manner. This can be used for UG and PG Zoology students, who are looking for a quick revision before the Practical Examination. This Allied practical book has been written to Students and preparatory the field of Life sciences, No such Manual which cover suitable original picture with needed explanation of basic Non-chordate, Chordates and General Principles of Zoology techniques and Practice.



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Kalaimani Subramani
Shenkani Krishnasamy

The Allied Practical Mannual

Non- Chordata, Chordata and General Principles of
Zoology

LAP
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Publisher:

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str. A.Russo 15, of. 61, Chisinau-2068, Republic of Moldova Europe

Printed at: see last page

ISBN: 978-620-4-72938-1

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“THE ALLIED PRACTICAL MANNUAL
NON- CHORDATA, CHORDATA AND GENERAL PRINCIPLES
OF ZOOLOGY”

(A syllabus based on Periyar University)

FIRST EDITION

2021

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By

Mr. S. KALAIMANI, M.Sc., M.Ed., M.Phil, SET (Life Science), SET (E.du), D.C.A.,

ACKNOWLEDGEMENT

Words find no bounds to express my sincere thanks to all who made this Allied practical manual “THE ALLIED PRACTICAL MANUAL OF NON-CHORDATA, CHORDATA AND GENERAL PRINCIPLES OF ZOOLOGY” (a syllabus based on periyar university) to come true.

I owe my sincere thanks to Dr.SEERANGANAYAKI , Principal, J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal D.t, for extending the facilities to undergo Allied Practical Manual.

I thank Dr.S.UMAWATHY, Assistant Professor and Head, PG and Research Department of Zoology, J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal Dt.

My specific thanks to Dr.K.SHENKANI, Assistant Professor, PG and Research Department of Zoology, J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal Dt. for her kind co-operation and encouragement to complete this Allied practical Manual successfully.

I wish to express my profound thanks to All the Staff Members, Department of Zoology, J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal Dt.

I also wish to thanks Assistant Professor of the Department of Zoology, My heartfelt thanks to my Research supervisor Dr.C.Kandeepan, Assistant professor of zoology, APAC, palani for his cooperation and loving encouragement to complete this Allied practical Manual successfully.

My special thanks to Mrs. V.N.PADMA, Assistant CTO, for helping me in many ways for completing this Allied practical Manual work. With great pleasure I express my deep sense of feeling of my affectionate Father Late Mr. P.SUBRAMANI (late), Sub registrar, My mother Mrs. S.VEERAMMAL (late), My brother Mr. S.VELUMANI, Junior Asst Registration Department, My wife Mrs. V.N.PADMA, Assistant CTO, My Kids K.DEEPTHADHAYAZHINI and K.MITHUNMONISH, to whom I owe everything for their dedicated efforts and loving encouragement to complete this Book successfully.

And last but not least I wish to extend my sincere thanks to All My Friends and Relatives for their co- operation and support in this Allied Practical Manual to make success.

Mr. S. KALAIMANI

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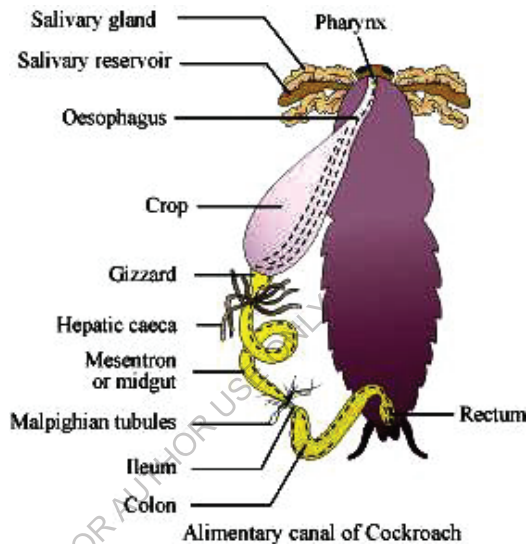
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UNIT: 1

DISSECTION

1.1 COCKROACH: ALIMENTARY CANALS



STOMODACUM:

- The fore gut consists of mouth, buccal cavy, pharynx, oesophagus, erop and glzzard. The mouth opens into a preoral cavity or buccal cavity or buccal cavity is situated the hypopharynx.
- The buccal cavity leads into a tubular pharynx which in turn passes vertically and then opens into a large thin walled pyriform rop.
- The crop extends upto the abdomen. The crop is posteriorly followed by a small thick walled, muscular pear shaped gizzard or proventriculus.

- Its inner wall is raised to form six hard cuticular teeth which are used as grinding apparatus.

MESENTERON:

- The mesenteron is a short narrow tubular structure and is lined by glandular epithelium.
- At the anterior region of the mid gut there are eight finger like blind mesenteric caecae or digestive caecae.
- These caecae secrete digestive fluid which is put into the mesenteron.

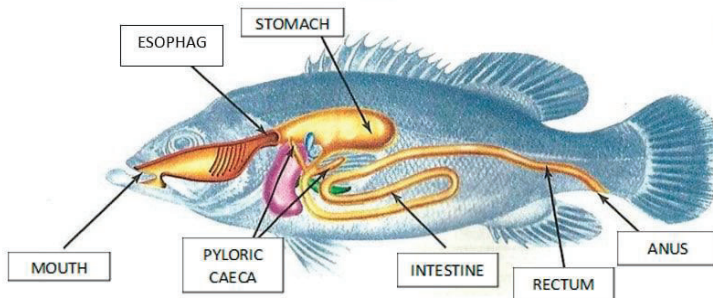
PROCTODAEUM:

- The hind gut is coiled and consists of a short narrow proximal ileum, a long coiled colon and a broad terminal rectum.
- The inner wall of the rectum is thrown into many longitudinal folds or rectal papillae.
- The rectum opens outside by anus.

SALIVARY GLANDS:

- There is a pair of salivary glands which are situated in the thorax on each side of the crop.
- Each gland consists of a bipartite glandular portion and a thin walled sac called the reservoir or receptacle.

2.2. FISH –ALIMENTORY CANALS



The structural components of a fish's digestive system include the mouth, teeth and gill rakers, esophagus, stomach, pylorus, pyloric caeca, pancreatic tissue (exocrine and endocrine), liver, gall bladder, intestine and anus.

A fish's digestive system is adapted to their food habits. In predatory (carnivorous) fishes, the mouth is usually large for engulfing prey whole, or in large chunks, and teeth are present on the jaws (*e.g.* maxillary and dentary) and tongue (*e.g.* glossyhyal) for grasping live prey.

Gill rakers are short in carnivorous fish and pharyngeal teeth are short and pointed for moving prey down the throat. In omnivorous and planktivorous fishes, the mouth is smaller and is usually devoid of teeth except for pharyngeal teeth that may be blunt and flat (molariform) for grinding

or sharp and long for shredding. Gill rakers in these fish are typically fine to prevent the escape across the gills of small food particles.

The esophagus, which is lubricated by mucus, leads to the stomach. In carnivorous fish, the stomach is muscular and elastic for holding large prey items, while in omnivorous and planktivorous fishes the stomach, if present at all, is small because a more or less constant stream of small food particles can flow directly into the intestine.

The pylorus is a sphincter that prevents premature movement of the food bolus out of the stomach. Around the pylorus, many fish have out-pocketings called pyloric caeca.

The role of pyloric caeca in digestion has been widely speculated upon. Histologically, they are very similar to the intestine. Around the pyloric caeca (or in that same area in fish lacking them) is pancreatic tissue. In fish the pancreas is usually diffuse, not a discrete body.

As in all vertebrates, the pancreas has two digestive functions. It is the source of: 1) exocrine secretion of digestive enzymes (proteases, lipases, and carbohydrases) into the intestine and 2) endocrine secretion of the hormones insulin and glucagon that act to lower and raise blood sugar.

The liver in fish produces bile which is stored in the gall bladder until a bolus passes the stomach, at which time the bile is expelled into the intestine. Bile

contains waste products of liver activity which pass out of the fish in the feces. Bile has a digestive function in that it emulsifies lipids, greatly increasing their absorption in the intestine.

The liver is key in the anabolism and catabolism of amino acids absorbed during digestion and is also the site of storage of food energy in the form of glycogen. Most absorption of nutrients occurs in the intestine. The large protein, fat, and starch molecules in food that have been broken into smaller molecules by gastric acid and digestive enzymes move by diffusion or active transport (ATPase pumps) into the network of capillaries surrounding the gut.

The intestine is lined with finger-like out-pocketings (villi) that greatly increase the surface area for absorption.

The intestinal wall, while not as active as the exocrine pancreas, can secrete digestive enzymes, as well.

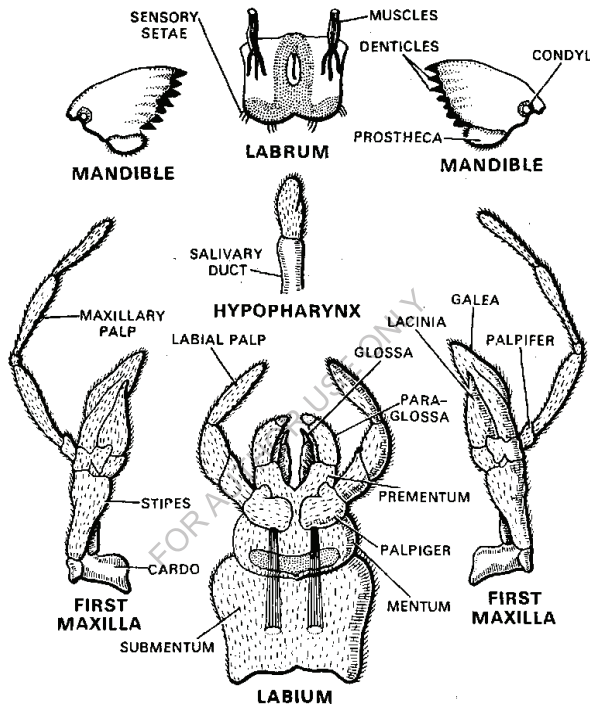
The alimentary canal is typically a short, S-shaped tube in carnivorous fish, but can be long and convoluted in omnivorous fish to aid in the absorption of less digestible plant material.

UNIT: 2

MOUNTINGS

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2.1. COCKROACH- MOUTH PARTS



- The mouthparts of cockroach are biting and chewing type. This biting and chewing type of mouthparts are considered as the most primitive and unspecialized of all the mouthpart types.

- The mouthparts of cockroach are developed to suit its habit of feeding on solid food and as a result it has well developed mandibles.
- These mouthparts help the cockroach to bite and chew on hard stuffs, consume soft stuffs and also lap upon liquids. The mouthparts also include Labrum, Mandibles, and a pair of first maxillae, labium, and hypopharynx.

Labrum:

- The mouth is covered by labrum. It is also known as upper lip. This labrum is attached to the clypeus.
- The labrum bears gustatory sensilla on its inner surface. Labrum helps in tasting and also handling the food.

Mandibles:

- These are a pair of triangular, hard, unjointed, stout, chitinated structures.
- The mandibles are located on either side of mouth behind labrum. They are dentate along their inner margins and are masticatory in function.

First pair of maxillae:

- A pair of first maxillae is located behind mandibles on either side of the mouth.
- The first maxilla has two basal segments called cardo and stipes.
- Cardo is attached to the head capsule and stipes is attached to the cardo.

- The stipes has five segmented maxillary palp on its outer side.
- This palp is situated on a small sclerite called palpifer. Inner to the palp two chitinous lobes namely lacinea and galea are found attached to stipes.
- Lacinea is pincer like with two terminal denticles whereas galea is the outer soft hood like structure bearing long chitinous bristles.
- The maxillary palps are used for cleaning the antennae and also the front pair of legs.

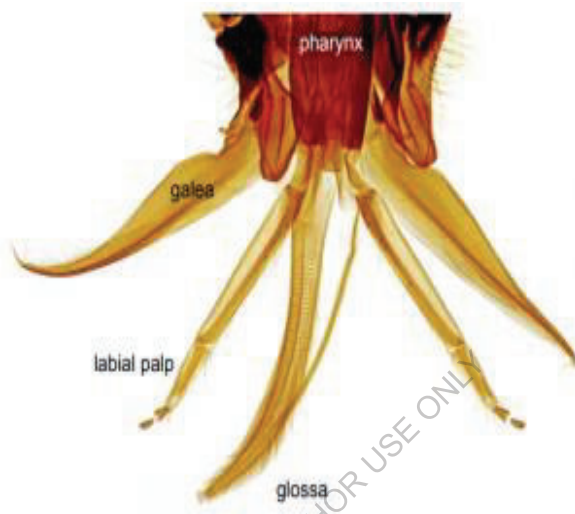
Labium:

- Labium is formed by the fusion of second pair of maxillae. It is also known as lower lip.
- The basal segment of labium is called post-mentum. Labium includes two segments namely broad rectangular sub-mentum and a triangular mentum.
- A pair of glossae is present between paraglossae. The paraglossae and glossae together constitute ligula.

Hypopharynx:

- It is chitinous, grooved and a rod-like structure found hanging into the preoral cavity.
- It is also known as ligula or tongue. Hypopharynx divides the proximal part of preoral cavity into a larger anterior cibarium and a posterior salivarium.
- The salivary duct opens into salivarium at the base of the hypopharynx.

2.2. HONEY BEE-MOUTH PARTS



- The mouthparts of honeybee are chewing and lapping type. Also bumble-bees also have similar kind of mouth parts.
- Honey bees have a combined mouth parts than can both chew and suck. This is accomplished by having both mandibles and a proboscis.

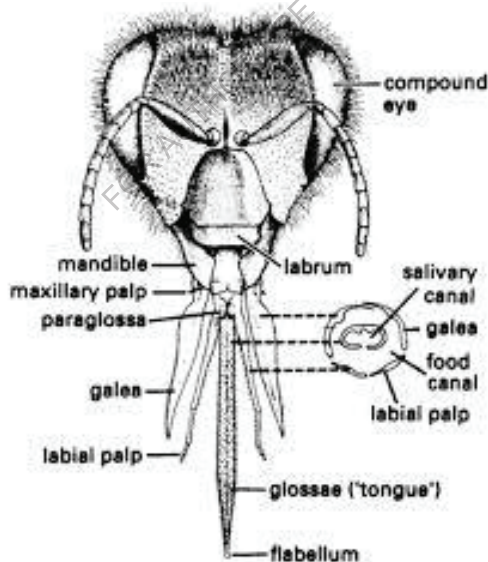
MANDIBLES:

- The mandibles are a pair of jaws suspended from the head of the bee. The insect uses them to chew wood when redesigning the hive entrance, to chew pollen and to work wax for comb-building.

- They also permit any activity requiring a pair of grasping instruments. These paired "teeth" that can be opened and closed to get the work done.

PROBOSCIS:

- The proboscis of the honeybee is not a permanent functional organ, but it is formed temporarily by assembling parts of the maxillae and the labium to produce a unique tube for drawing up liquids such as sweet juices, nectar, water and honey.
- The insect releases it when needed for use, then withdraws and folds it back beneath the head when it is not needed.



LABELLUM:

- The glossae are greatly elongated to form a hairy, flexible tongue.
- The glossa terminates into a small circular spoon shaped lobe called labellum, which is useful to lick the nectar.
- Labial palps are elongate and four segmented.

MAXILLOLABIAL STRUCTURES:

- Maxillolabial Structures are modified to form the lapping tongue.
- The tongue unit consists of the two galeae of maxillae, two labial Palps and an elongated flexible hairy glossa of labium.

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2.3. TYPES OF SCALES IN FISHES



Placoid



Ctenoid



Cycloid



Ganoid

- Take a cartilaginous fish in dissecting tray and cut open the animal with the help of scissor.
- Remove the skin, cut it into small pieces and put them in petridish.
- Take 1-2 pieces of skin in a watch glass having water and remove attached muscles with the help of forceps.
- Transfer the skin pieces into a test tube and add 2-3 ml of 10 % KOH with pipette.
- Gently boil it for few minutes preferably for 2-3 mins.

- Allow the test tube to stand for a few minutes. Decant the supernatant leaving a few drops in the test tube.

Pour this residual KOH into a watch glass.

- Place a drop from it onto a slide and put the cover slip without any air bubble.
- Observe under low power (10X objective) of a compound microscope.

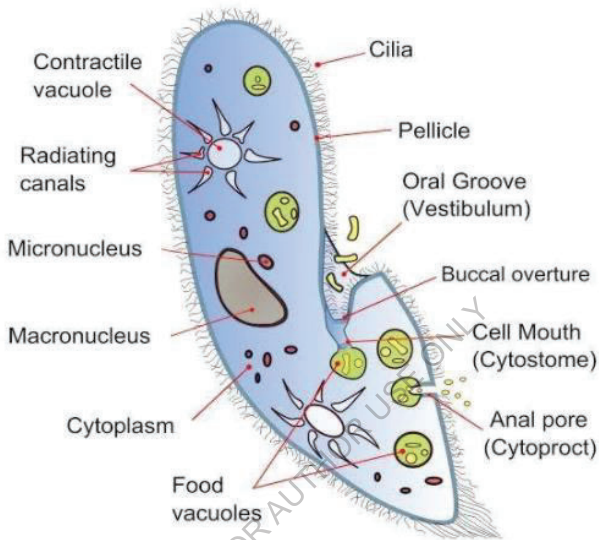
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UNIT: 3

SPOTTERS

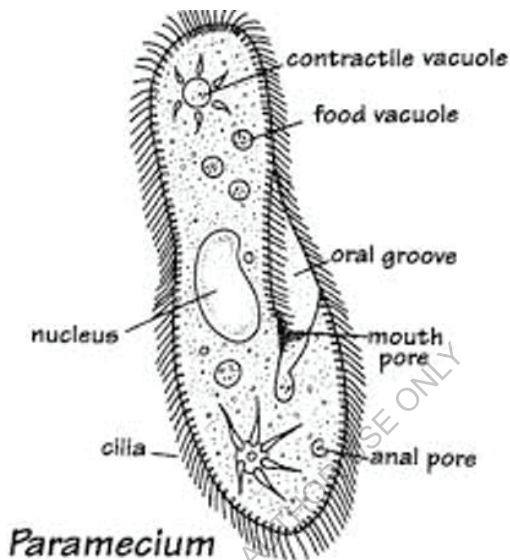
IDENTIFICATION

3.1. PARAMECIUM ENTIRE



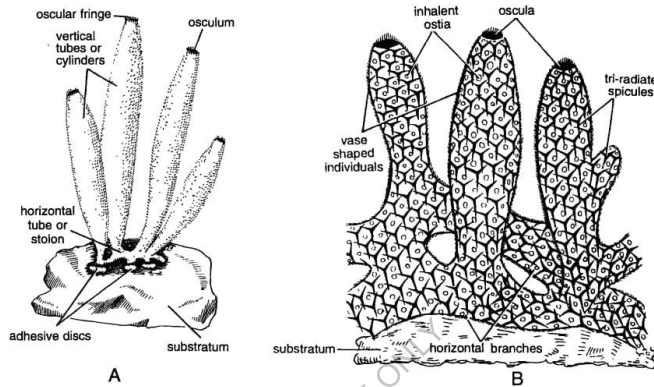
- Species of *Paramecium* range in size from 50 to 330 micrometers (0.0020 to 0.0130 in) in length.
- Cells are typically ovoid, elongate, and foot- or cigar-shaped.
- The body of the cell is enclosed by a stiff but elastic structure called the pellicle.
- This consists of the outer cell membrane (plasma membrane), a layer of flattened membrane-bound sacs called *alveoli*, and an inner membrane called the *epiplasm*.

- The pellicle is not smooth, but textured with hexagonal or rectangular depressions. Each of these polygons is perforated by a central aperture through which a single cilium projects.
- Between the alveolar sacs of the pellicle, most species of *Paramecium* have closely spaced spindle-shaped trichocysts, explosive organelles that discharge thin, non-toxic filaments, often used for defensive purposes.
- Typically, an anal pore (cytoproct) is located on the ventral surface, in the posterior half of the cell. In all species, there is a deep oral groove running from the anterior of the cell to its midpoint.
- This is lined with inconspicuous cilia which beat continuously, drawing food inside the cell. *Paramecia* live mainly by heterotrophy, feeding on bacteria and other small organisms.
- A few species are mixotrophs, deriving some nutrients from endosymbiotic algae (chlorella) carried in the cytoplasm of the cell.
- Osmoregulation is carried out by contractile vacuoles, which actively expel water from the cell to compensate for fluid absorbed by osmosis from its surroundings. The number of contractile vacuoles varies from one, to many, depending on species.
- in which *Paramecium* of compatible mating types fuse temporarily and exchange genetic material.



- A *Paramecium* propels itself by whiplash movements of the cilia, which are arranged in tightly spaced rows around the outside of the body.
- *Paramecia* feed on microorganisms like bacteria, algae, and yeasts
- *Paramecium* reproduces asexually, by binary fission. Under certain conditions, it may be preceded by self-fertilization (autogamy), it may follow conjugation, a sexual phenomenon

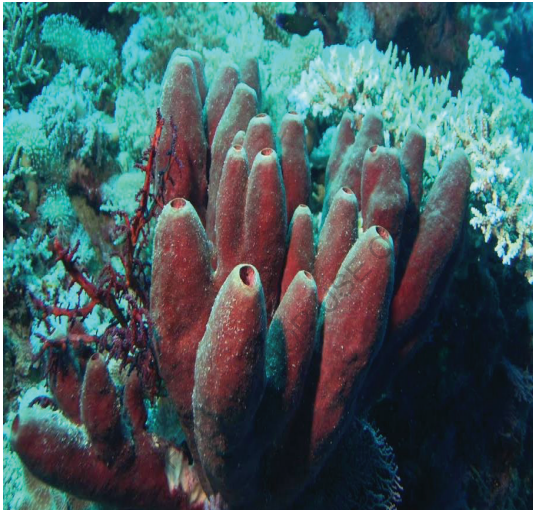
3.2. ASCON: ENTIRE



Leucosolenia. A. Colony, B. Portion of colony (magnified).

- Ascon sponges are the simplest and least common sponge body form.
- These sponges have multiple *ostia*, this is where water enters the sponge, and only have one *osculum*, where water exits the sponge.
- The openings lead to the sponge's one central chamber called a *spongocoel* that is lined with choanocytes.
- They are vase like in shape.
- Ascon sponges are the simplest and least common sponge body form.
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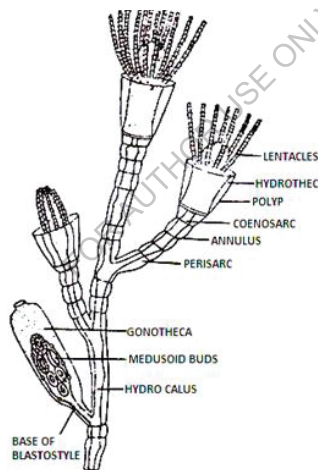


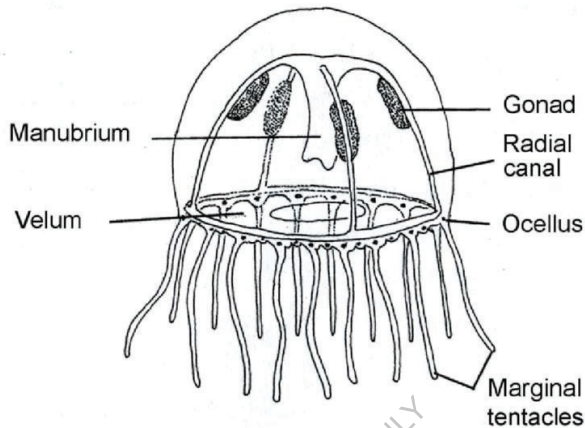
3.3. OBELIA COLONY MEDUSA



- Obelia is also called as Sea fur.
- The hydroid colony of Obelia is delicate, semitransparent and whitish to light brown in color.
- It consists of vertical branching stems are called as hydrocauli and the root like branches are called hydrorhiza. Both are of same thickness.
- The growth of the colony is sympodial.
- Each of the vertical stem or hydrocauli branches in an alternate manner.

- The ultimate branch terminates in a nutritive zooid called polyp or hydranth.
- In the older polyps cylindrical reproductive zooids are placed which is known as blastostyle or gonangia.
- This Obelia colony is dimorphic exhibiting two types of zooids. When the blastostyles develop saucer-shaped bodies called as medusae, the dimorphic colony becomes trimorphic.





MEDUSAE:

- These are small, transparent, solitary, free swimming saucer-shaped or bell-shaped zooids.
- They measure about 6 mm in diameter. These are the reproductive zooids which produce the sex cells.
- The inner concave side of the body is known as sub-umbrella and outer convex as exumbrella.
- A short, hollow, quadrangular projection, the manubrium, hangs down from the middle of the sub umbrella surface.
- This structure together with the disc or bell-shaped body of the medusa gives it an umbrella-like appearance.
- The mouth, a square or four sided opening is situated at the tip of the manubrium.

- The mouth leads through the cavity of the manubrium into a small gastral cavity or enteric cavity or stomach situated in the central part of the main body of the medusa.
- From this cavity radiate four narrow radial canals situated at equal distances from each other.
- These canals run outwards toward the edge of the umbrella and open into a circular canal running around the edge of the umbrella.
- This system of canal enables the food to be taken in at the mouth and manubrium and digested in the stomach to be distributed through them to the entire medusa.
- The edge of the medusa gives off on its inner side a very narrow, rudimentary fold or shelf called the velum.
- The margin of the umbrella gives off short tentacles which are sixteen in number in the newly born medusa but are numerous in the adult.
- Ectoderm covers both the surface of the umbrella and the outer surface of the manubrium.
- The endoderm lines the cavity of the manubrium, 'stomach', radial canals and circular canals which together represent the enteron. Endoderm does not extend into the velum.

3.4. FASCIOLA HEPATICA: ENTIRE



- *Fasciola hepatica*, also known as the common liver fluke or sheep liver fluke, is a parasitic trematode (fluke or flatworm, a type of helminth) of the class Trematoda, phylum Platyhelminthes.
- It infects the livers of various mammals, including humans, and is transmitted by sheep and cattle to humans the world over.
- The disease caused by the fluke is called fasciolosis or fascioliasis, which is a type of helminthiasis and has been classified as a neglected tropical disease.
- Fasciolosis is currently classified as a plant/food-borne trematode infection, often acquired through eating the parasite's metacercariae encysted on plants.

- *F. hepatica*, which is distributed worldwide, has been known as an important parasite of sheep and cattle for decades and causes significant economic losses in these livestock species, up to £23 million in the UK alone.
- Because of its relatively large size and economic importance, it has been the subject of many scientific investigations and may be the best-known of any trematode species.
- *F. hepatica*'s closest relative is *Fasciola gigantica*. These two flukes are sister species; they share many morphological features and can mate with each other.

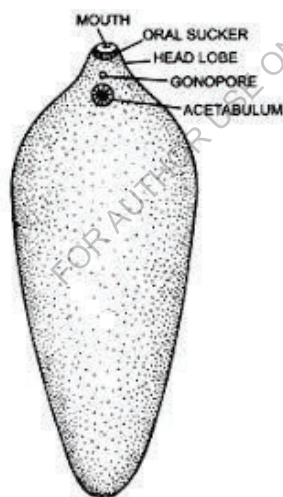


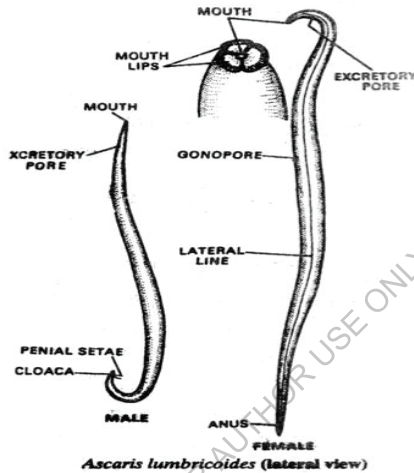
Fig. 5.6. *Fasciola hepatica*.

3.5. ASCARIS: ENTIRE



- *Ascaris* is a genus of parasitic nematode worms known as the "small intestinal roundworms", which is a type of parasitic worm.
- One species, *Ascaris lumbricoides*, affects humans and causes the disease ascariasis.
- Another species, *Ascaris suum*, typically infects pigs. *Parascaris equorum*, the equine roundworm, is also commonly called an "ascarid".
- Their eggs are deposited in feces and soil. Plants with the eggs on them infect any organism that consumes them.
- *A. lumbricoides* is the largest intestinal roundworm and is the most common helminth infection of humans worldwide.
- Infestation can cause morbidity by compromising nutritional status, affecting cognitive processes, inducing tissue reactions such as granuloma to larval stages, and by causing intestinal obstruction, which can be fatal. Adult: cylindrical shape, creamy white or pinkish in color.

- Male: average 15–30 cm (6–12 inches); more slender than the female
- Female: average 20–35 cm (8–14 inches)

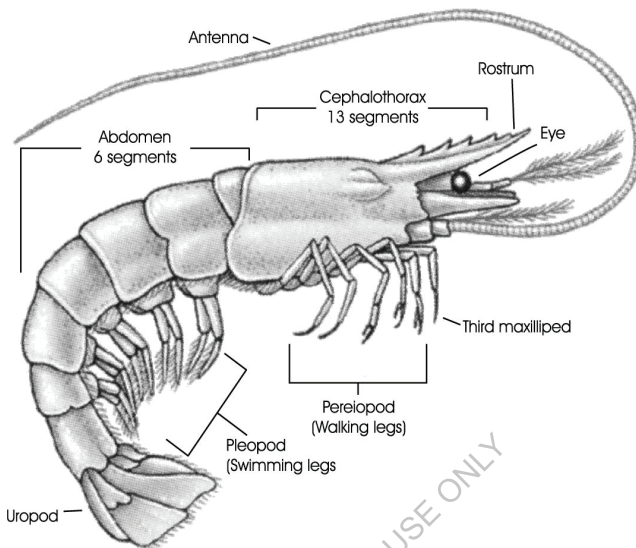


- The body is long, cylindrical, and fusiform (pointed at both the ends).
- The body wall is composed of cuticle, epidermis and musculature.
- There is a pseudocoelom.
- Respiration is by simple diffusion.
- The nervous system consists of a nerve ring and many longitudinal nerve cords.
- Reproduction is exclusively sexual, and males are usually shorter than females.

3.6. PENAEUS: ENTIRE



- The giant tiger prawn inhabits the coasts of Australia, South East Asia, South Asia and East Africa.
- Similar to all penaeid shrimp, the rostrum well developed and toothed dorsally and ventrally.
- Carapace without longitudinal or transverse sutures. Cervical and orbito-antennal sulci and antennal carinae always present.
- Hepatic and antennal spines pronounced. Pterygostomian angle round. Stylocerite at first antennular segment.
- Basal spines on first and second pereopods and exopods on the first to fourth pereopods usually present.
- No fixed subapical spines on telson. Adrostral sulcus and carina are short, not reaching posteriorly beyond midlength of carapace.
- Gastrofrontal carina absent. Females have closed-type thelycum. Petasma in male symmetrical with thin median lobes.

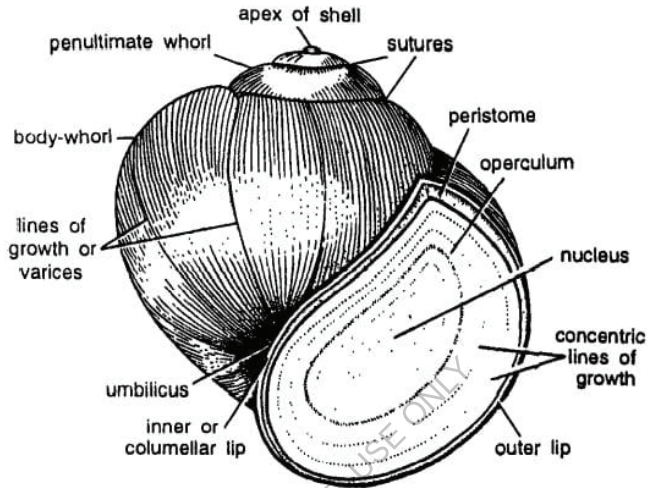


- The most distinct features for identification of this species are: fifth pereopods without exopod; hepatic carina horizontally straight; and gastroorbital carina occupying the posterior half of the distance between hepatic spine and postorbital margin of carapace.
- Depending on substratum, feed and water turbidity, body colours vary from green, brown, red, grey, blue and transverse band colours on abdomen and carapace are alternated between blue or black and yellow.
- Adults may reach 33 cm in length and females are commonly larger than males.

3.7. PILA: ENTIRE



- *Pila* is a genus of large freshwater snails with an operculum, African and Asian apple snails, aquatic gastropod mollusks in the family Ampullariidae, the apple snails
- Distribution of the genus *Pila* include Africa, Madagascar, southern Asia and Indo-Pacific islands.
- Planktonic development occurs in *pila* from adult stage to its death
- *Pila* species are a host of a trematode *Multicotyle purvisi*.



- The shells of *Pila* are used in traditional ethnomedicine for weakness by Saharia people in Rajasthan, India.
- *Pila ampullacea* and *Pila pesmei* are some of the rice field snail species traditionally eaten in Thailand that have been displaced by the invasive golden apple snail, *Pomacea canaliculata*.

3.8. STAR FISH ENTIRE-ORAL AND ABORAL VIEW

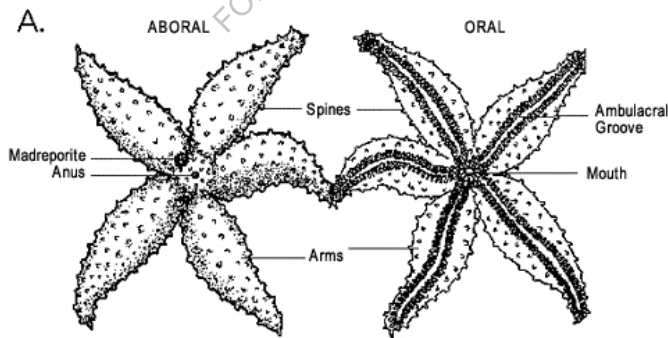


- Starfish or sea stars are star-shaped echinoderms belonging to the class Asteroidea.
- Common usage frequently finds these names being also applied to ophiuroids, which are correctly referred to as brittle stars or basket stars.
- Starfish are marine invertebrates. They typically have a central disc and usually five arms, though some species have a larger number of arms.
- The aboral or upper surface may be smooth, granular or spiny, and is covered with overlapping plates.
- Many species are brightly coloured in various shades of red or orange, while others are blue, grey or brown.
- Starfish have tube feet operated by a hydraulic system and a mouth at the centre of the oral or lower surface. They are opportunistic feeders and are mostly predators on benthic invertebrates.
- Several species have specialized feeding behaviours including eversion of their stomachs and suspension feeding.

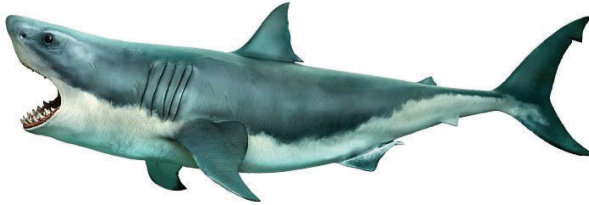
- They have complex life cycles and can reproduce both sexually and asexually. Most can regenerate damaged parts or lost arms and they can shed arms as a means of defense.
- The Asteroidea occupy several significant ecological roles. Starfish, such as the ochre sea star (*Pisaster ochraceus*) and the reef sea star (*Stichaster australis*), have become widely known as examples of the keystone species concept in ecology.
- The tropical crown-of-thorns starfish (*Acanthaster planci*) is a voracious predator of coral throughout the Indo-Pacific region, and the northern Pacific sea star is considered to be one of the world's 100 worst invasive species.

Distribution:-

Asterias is found in shallow water in North Temperate seas and found abundantly on North-Atlantic coast. It is found in abundance in India and U.S.A.



3.9. SHARK ENTIRE



- Sharks are a group of elasmobranch fish characterized by a cartilaginous skeleton, five to seven gill slits on the sides of the head, and pectoral fins that are not fused to the head.
- Morphology of a shark: large, long and very powerful selachian fish.
- It is viviparous.
- Head: foremost part of a shark, comprising the skull and face.
First dorsal fin: front locomotive limb on the back of a shark.
Second dorsal fin: rear locomotive limb on the back of a shark.
Caudal fin: locomotive limb at the end of a shark.

Pelvic fin: locomotive limb beneath the pelvic girdle of a shark.

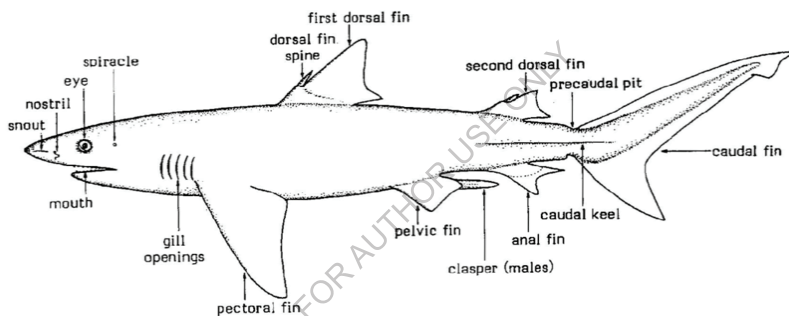
Belly: lower part of the trunk of a shark.

Pectoral fin: locomotive limb on the chest of a shark.

External gill openings: gill slit.

Mouth: entrance to the digestive tract of a shark.

Eye: sight organ of a shark.



3.10. FROG: ENTIRE



- An adult frog has a stout body which is differentiated into head and trunk. Other external features are a pair of nostrils, protruding eyes, a membranous tympanum (ear), slippery/warty moist skin and webbed limbs.
- Frogs generally have a slippery moist and highly permeable skin through which they absorb water and respire. the moist skin acts as a respiratory organ in frogs. Also, the skin is glandular in nature, which produces mucus and toxic substances to warn them of their predators.
- The colour of the skin can vary from brown and green to vivid colours as per secretions.
- The locomotion of frogs takes place with the help of their forelimbs and hind limbs.

- Frogs are unisexual i.e., they show sexual dimorphism. A male frog is distinguished from a female frog by the presence of vocal sacs and a copulatory pad on forelimbs.
- A female frog lacks these body features.

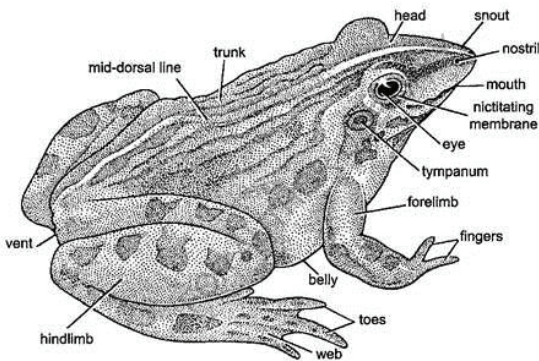


Fig. 18.3. An Indian frog, *Rana tigrina*, in sitting posture.

3.11. CALOTUES: ENTIRE



SHAPE, SIZE AND COLOUR:

- Body is elongated, slender, lizard-like and divisible into head, neck, trunk and tail. Male is brightly coloured, golden yellow with a greenish tinge. Throat is scarlet red, often with a black transverse bar.
- Female is not so brightly coloured. Male is larger and stronger than female, measuring about 35 cm in length including tail

HEAD:

- The head is large and somewhat pyramidal in shape with a short conical snout which bears a wide, slit-like aperture, the mouth running round the anterior border of the head and a pair of small external nares situated above the anterior extremity.

- At the sides of the head are the eyes, each provided with upper and lower opaque, movable eyelids and with a transparent third eyelid or nictitating membrane, which, when withdrawn, lies in the anterior angle of the orbit.
- Behind the eye is a circular patch of skin-the tympanic membrane somewhat sunk below the general level of the skin.

TRUNK:

- Head is connected with the trunk with a short neck. The trunk is elongated, strongly convex dorsally, laterally compressed and flattened ventrally.
- At the junction of trunk and tail, on the ventral surface is a slit-like transverse aperture, the cloacal aperture.
- The trunk bears two pairs of limbs. Each forelimb has an upper-arm or brachium, the forearm or antebrachium, the hand or manus and five horny clawed digits; while each hindlimb comprises the thigh or femur, shank or crus, foot or pes and five clawed digits, of which the hallux or first is the smallest.
- Tail is cylindrical, tapering showing alternate dark and light annuli and is nearly twice as long as the head and trunk together.
- In adult male the base of tail just behind the cloacal aperture becomes quite swollen due to hemipenes.

SKIN AND EXOSKELETON:

- Skin is dry and devoid of glands. The body is entirely covered by epidermal horny, imbricate and backwardly directed scales of unequal size which are periodically shed in flakes.
- On the head, a prominent shield covers the parietal foramen and two spines are present above each ear opening. Over the head and trunk, along the mid-dorsal line, is a characteristic crest of large spine-like movable scales directed backward and gradually diminishing in size posteriorly. here are melanophores lying deep in the dermis and iridocytes and lipophores more superficially.
- It is able to change colour. The colour may change with the environment, serving the purpose of concealment.
- Melanophores are controlled partly or entirely by the sympathetic nervous system. Colour is largely controlled through the eyes. Histology of skin is similar to that of Uromastix.

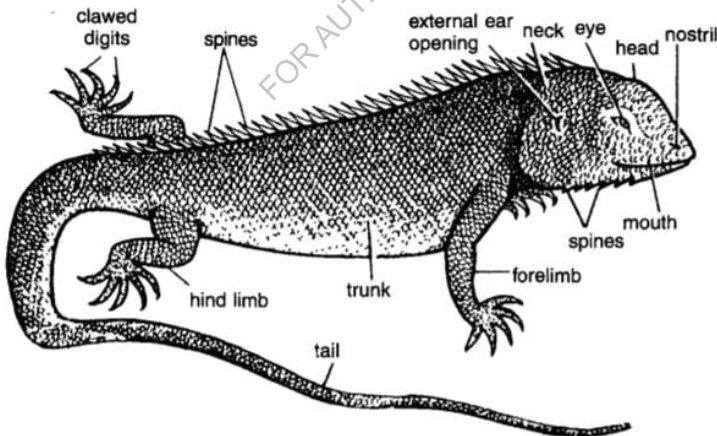
SCALES:

- Scales form the continuous covering over the body of Garden Lizard (Calotes), but become thinner in the grooves between the scales.
- Scales on the body of Calotes are of two types- large scales and small scales. Sensory structures in the form of bristles are present on both the scales called prototrichus.

- Scales are epidermal in origin, develop from Malpighian layer. Each scale is a plate-like structure supported by a bony plate or the ossicle.

CLAWS:

- Tips of digits of Garden Lizard (*Calotes*) are provided with sharp claws. Each claw is made up of a dorsal and ventral scale-like horny plate.
- Dorsal plate is called unguis and ventral plate is called subunguis. The latter is flattened. Both these plates are derived from the Malpighian layer and placed in such a way that their coverage should form a sharp claw.
- Cellular structure of skin is similar to that of *Uromastix*.



3.12. PIGEON: ENTIRE



SHAPE AND SIZE:

- The spindle-shaped or fusiform body of Blue Rock Pigeon is about 33 cm in length and is well adapted for rapid movement through the air.

COLORATION:

- Except the eyes and the feet which are pink, rest of the body of pigeon is a salty gray with glistening metallic green and purple sheen on the upper breast and around the neck. The wing has two black bars.

.1. Head:

The head is small, rounded and mobile.

(i) Beak:

- Head is prolonged in front into a short, pointed, bill or beak, which is formed by an elongated upper jaw and comparatively smaller lower jaw.

Both jaws lack teeth and remain ensheathed by a horny sheath or rhamphotheca.

- The terminal mouth is thus guarded by horny beaks and forms a wide gape.

(ii) Nostrils:

- At the base of upper beak occurs a patch of naked, whitish, swollen area of soft skin, called operculum or cere.
- The cere overhangs on two oblique, slit-like external nares or nostrils which can be closed by the cere.

(iii) Eyes:

- On either lateral side of the head is located a large eye. The eyes are round and are guarded by upper and lower eyelids and a transparent third eyelid or nictitating membrane.
- The nictitating membrane can be drawn across the eyeball from the anterior upper corner of the eye. The eyelids lack eyelashes and eyebrows,

(iv) Ear apertures:

- Below and behind each eye is an external ear opening which leads to a short tube, the external auditory meatus, closed by the tympanic membrane.
- These apertures are hidden under special auricular feathers.

2. Neck:

- The neck is long, flexible and well demarcated from head and trunk.

- It helps in handling of food and compensates the forelimbs which have modified into wings.

3. Trunk:

- The trunk is compact and spindle-shaped, bearing a prominent bony ridge, the keel or carina, at its mid-ventral line.
- There occurs a large, transversely elongated cloacal aperture on the ventral surface at the function of the uropygium (tail) with the trunk.
- The trunk bears two pairs of pentadactyl type limbs- one pair of wings and one pair of legs.

(i) Wings:

- In pigeons, two forelimbs are modified into two powerful organs of flight, the wings. Each wing remains attached high on the antero-dorsal side of the trunk. Each wing is elongated, flattened and distally pointed with its longitudinal axis at right angles to that of the trunk.
- It can be folded close to the body in the form of Z during rest and extended during flight. The wings show the three typical divisions of the forelimb- upper-arm or brachium, forearm or ante-brachium and the hand or manus, all closely bound together by skin.
- The manus or hand bears three, imperfectly-marked and clawless digits, of which the second is the largest.

- On the anterior or pre-axial border of the limb a fold of skin stretches between the upper-arm and the forearm called the alar membrane or pre-patagium.
- A similar but much smaller fold extends, post-axially, between the proximal portion of the upper-arm and the trunk, called the post-patagium.

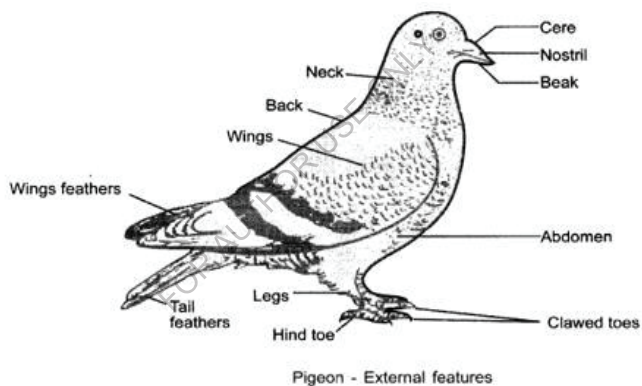
(ii) Legs:

- The hindlimbs or legs arise somewhat anteriorly from the trunk to balance and support the entire weight of the body at rest. Each hindleg comprises three parts: the thigh, shank and foot.
- The thigh is short, closely bound to the trunk, and directed downwards and forwards. The shank is long and extends from the knee downwards and backwards.
- The foot is clearly divisible into a proximal portion, the traso-metatarsus, and four digits. The digits are long and clawed.
- The first digit is called hallux and is directed backwards. The remaining three digits, the 2nd, 3rd and 4th, extend forward. The legs are efficient organs of perching and locomotion with long toes and curved claws.

4. Tail:

- After removal of the feathers it can be seen that the true tail is short, conical projection of the trunk, known as uropygium. It bears a group of elongated tail-feathers or rectrices.

- On the dorsal surface of uropygium is a papilla bearing on its summit the opening of a preen, coccygeal, or oil-gland.
- It is assumed that the vitamin would be ingested during preening. It is also probably used for keeping the horny beaks from becoming brittle.
- The tail with its feathers is used as a rudder during flight and a balancer in perching.



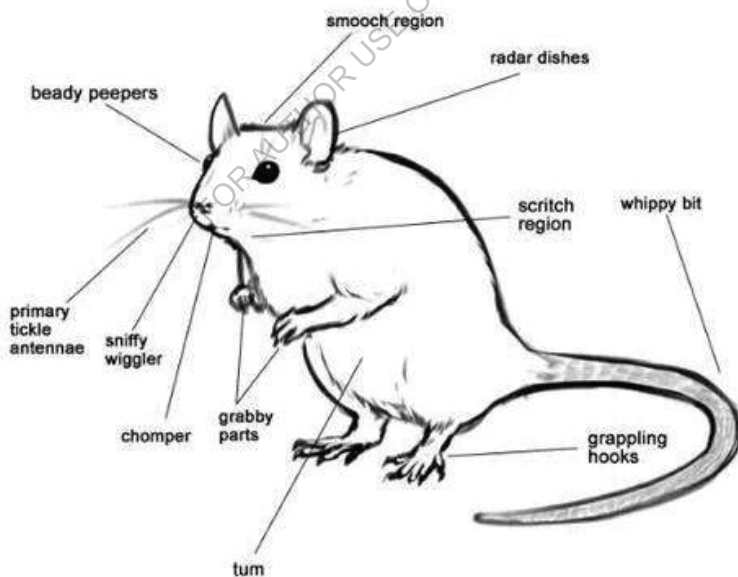
3.13. RAT: ENTIRE



- Rats are generally slender with a pointed head, large eyes, and prominent, thinly furred ears.
- They have moderately long legs and long, sharp claws. The bald soles of their narrow hind feet possess fleshy pads of variable size, depending on species.
- The brown rat has a larger body than the house rat, and its tail is shorter relative to the body.
- The brown rat also has thicker fur and 12 pairs of mammae instead of 10. Tail length among rats ranges from shorter than body length to appreciably longer. The tail appears smooth and bald but is actually covered with very short, fine hairs.
- In a very few species, these hairs become longer toward the tip, which gives the tail a slightly tufted appearance.
- As with any large group of rodents, body size varies within the genus. Most species are about the size of Hoffman's rat (*R. hoffmanni*), native to the

Indonesian island of Sulawesi and weighing 95 to 240 grams (3.4 to 8.5 ounces), with a body length of 17 to 21 cm (6.7 to 8.3 inches) and a tail about as long.

- One of the smaller species is Osgood's rat (*R. osgoodi*) of southern Vietnam, with a body 12 to 17 cm long and a somewhat shorter tail.
- At the larger extreme is the Sulawesian white-tailed rat (*R. xanthurus*), measuring 19 to 27 cm long with a tail of 26 to 34 cm.

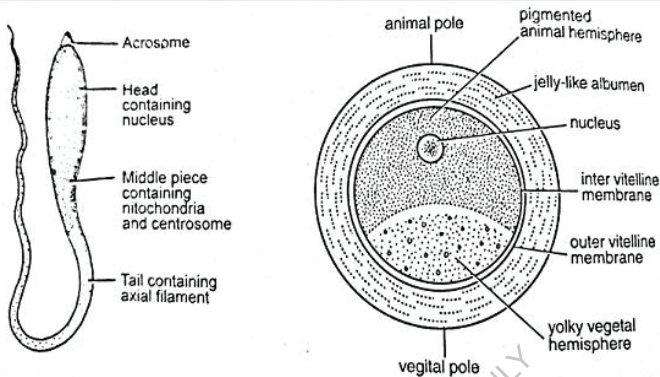


UNIT: 4

EMBRYOLOGY

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4.1. STRUCTURE OF EGG AND SPERM OF FROG



- Sperm of frogs that are not released in water like the foam nest frog and the Cape Rain frog have very long sperm heads of over 20 μm in length and show either no motility or very poor sperm swimming ability compared to the toad, *Bufo*, with shorter sperm heads which swims fast for frogs and about the same speed as human sperm as analysed by CASA.
- Sperm motility in frogs can be studied by means of SCA CASA and have many applications both in terms of artificial fertilization and sperm competition and assessing motility after cryo-protection.
- Nigrosin – eosin based stain such as BrightVit also distinguishes well between live and dead sperm.

- **Egg** The egg for frog is about 2mm in diameter at the time of ovulation. It is surrounded by two accessory egg membranes in addition to the plasma membrane.
- Just outside the plasma membrane is a non living transparent membrane called vitelline membrane developed by the 2 ovum itself.
- Outer to vitelline membrane is the jelly coat or albumen secreted by the walls of the oviduct.
- As soon as the egg reaches the water, the jelly coat swells up by the imbibitions of water and it protects the egg from injury and against infection by bacteria and other microorganisms.
- Frog's egg exhibits a well developed polarity and radial symmetry.
- The cytoplasm has two regions, the cortex and endoplasm. 1. Egg cortex A jelly like viscous layer of cytoplasm adherent to the plasma membrane is called ectoplasm or egg cortex.
- It possesses some membrane bound spherical bodies called cortical granules containing acid mucopolysaccharides. These remain arranged in a layer close to the plasma membrane.
- Dark-brown pigment granules are present in the egg cortex on the animal hemisphere. The presence of these granules imparts a dark brown colour to the entire animal hemisphere.
- The vegetal pole is whitish with little pigment granules. The cortical layer of egg is stable and is not shifted by streaming movement of cytoplasm or centrifugation force.

- It plays an important role in the development of egg. The egg cortex is responsible for establishing polarity, bilateral symmetry and general organization of the developing egg.
- Endoplasm The inner ooplasm with its nucleus is called endoplasm which is colloidal in nature. This portion contains cell organellae like mitochondria and ribosomes, and also organic and inorganic substances.
- Endoplasm contains a cup shaped mass of white yolk platelets called vitelline cupola. The germinal vesicle or nucleus is located near the animal pole. The yolk granules are little and small sized in the animal pole while they are heavily deposited in the vegetal pole,
- Frog's egg is said to be mesolecithal and moderately telolecithal since it contains a moderate amount of yolk which is distributed unevenly in the cytoplasm, the vegetal pole having the highest concentration.



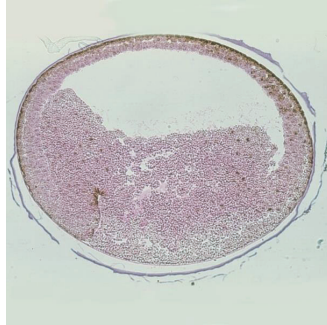
4.2. BLASTULA OF FROG



- At the end of cleavage the solid ball of cells give rise to blastula which consists of a number blastomeres.
- The characteristic features of the blastula stage are the presence of a well defined cavity called the blastocoel.
- This is the beginning of the primary body cavity. The process of the formation of blastula is called blastulation.
- The blastula of frog is called amphiblastian as the cavity is confined to only the animal pole.
- The vegetal pole however is composed of a solid mass of non pigmented yolky cells. In the thirty two cell stage, the blastula consists of a single layer of cells and is called the early blastula.

- The pigmented cells (micromeres) are found in the anterior half while the yolky megameres are present in the posterior half. As has been already pointed out, the blastocoel lies entirely in the anterior half.
- The blastula of frog is hollow and has a very well developed blastocoel. It is said to be a coeloblastula. As segmentation proceeds, the number of cells in the blastula increase; so also the blastocoel.
- The floor of the blastocoel is flat while its top portion is arched. The roof is made up of three to four layers of pigmented micromeres while the floor is formed by yolky megameres. Between the micromeres and the megameres and along the equator is found a group of cells which are intermediate in size (between megameres and micromeres).
- These cells constitute the germ ring. The germ ring is formed in the region of the grey crescent.

4.3. GASTRULA OF FROG

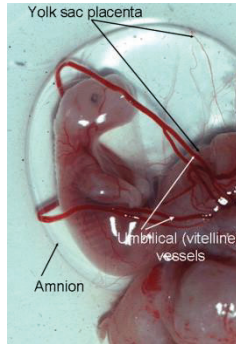


- At the end of gastrulation, we end up with a three-layered frog embryo.
- These three layers are the germ layers, which are the basis for all of the organs that will form later on.
- The outermost layer of the embryo is the ectoderm. This layer will become many different linings but most superficially will become the outer layer of the skin, part of the esophagus, as well as many other organs and nerve tissue.
- On the inside of the gastrula is the endoderm, which forms the digestive tract of the embryo.
- The endoderm will become the digestive tract of the organism as well as a number of other assorted organs and linings, such as the lungs, liver, etc.

- The middle layer is the mesoderm. This ends up becoming many of the skeletal muscles of the organism. This also becomes the bone, blood, and sex organs.
- This is just a quick overview of the three germ layers as they give rise to many different organs and organ systems. It's also after gastrulation, when the organs form, that we see some of the major differences between organisms as they develop.

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4.4. PLACENTA OF MAMMALS-RABBIT



- The egg of rabbit after fertilization reaches the uterus which consists of two long, tubular “horns”.
- A single cleaving egg or blastocyst generally lodges itself in one of the crypts at the midpoint of the horn containing it.
- If two or more cleaving eggs are present, they become evenly spaced along the horn. To prevent the sticking of the blastocyst with any other maternal tissue, the zona pellucida remains attached to it, till it reaches the uterine cavity.

Process of Placentation:

- After the three primary germ layers are laid down in the embryo, remarkable physiological changes take place in the uterine wall of the mother.
- The region where the uterine wall comes in contact with the embryo, bleeds due to some enzymatic action.
- The process of implantation in rabbit is initiated and controlled by a protein called blastokinin which is secreted by the uterine wall.
- The maternal site of implantation subsequently becomes thick, spongy and vascularized. The superficial contact is made intimate by the formation of minute fingerlike projections called villi
- The villi develop as an outgrowth from the extra embryonic region of the foetus and penetrate into the depressions present in the wall of the uterus.
- These villi are initially formed by the trophoblast. Later the connective tissue and blood vessels enter into it. At this time they are designated as “chorionic villi” and the blood vessels that enter into it are the ramifications of the allantoic blood vessels.
- The portion of the uterine wall that has undergone vascularization is called trophospongia.
- The villi of the trophoblast penetrate deep into the trophospongia and ramify extensively into it, establishing intimate physiological relationships.

- During this action, the villi are constantly bathed by the maternal blood due to excessive vascular state of the trophospongia.
- The villi develop as an outgrowth from the extra embryonic region of the foetus and penetrate into the depressions present in the wall of the uterus. These villi are initially formed by the trophoblast.
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UNIT: 5

**ADAPTATIONAL
SIGNIFICANCE**

5.1. NATATORIAL-TURTLE



- Turtle: Spends most of its life in the water.
- Turtles tend to have webbed feet for swimming. Sea turtles (Cheloniidae family) are especially adapted for an aquatic life, with long feet that form flippers and a streamlined body shape.
- They rarely leave the ocean, except when the females come ashore to lay their eggs, although some, such as the green sea turtle, do come out on reefs and beaches to bask.
- Other turtles live in fresh water, like ponds and lakes.
- They swim, but they also climb out onto banks, logs, or rocks to bask in the sun.
- In cold weather, they may burrow into the mud, where they go into torpor until spring brings warm weather again.

5.2. VOLANT-DRACO AND BAT



CHARACTERISTICS OF VOLANT ATMOSPHERE:

- The environment is characteristically homogeneous and lighter than any other habitable places.
- There is less possibility of friction with environment during movement. The only problem is the gravitational force of earth.

Pre-requisites for Volant Life:

- Development of specialized organs for flying is the primary requirement. Body of a volant animal must be light and rigid.
- During flight continuous supply of energy and power to the muscles is necessary. Animals are also required to acquire speed adaptation. Moreover, balancing and navigation of the body during flight should be well integrated.

Adaptive Features:

- The volant adapted animals developed flight mechanism as their prime instrument to overcome the attraction force of earth.

There are two types of flight that can be observed in volant animals:

(i) Passive and

i. Passive Flight:

- In this type of flight, animals jump from a high point, like hilltop, treetop etc. and come down to earth due to the attraction force of gravity.
- During their downward journey they cover a horizontal distance with the help of some sustaining organs.
- In this type of flight the animal never exerts any locomotory force, instead they move passively.

There are two types of passive flight among the animals:

(a) Parachuting and

(b) Gliding.

(a) Parachuting:

- The animal during passive flight when comes down on the earth at an angle of 45° or more then it is called parachuting. If we consider the angle with

respect to the point of jumping then it can be seen that they are travelling less distance than the gliders

(b) Gliding:

- When the passive flier comes down on the earth at an angle less than 45° then it is called gliding.
- This type of flier covers more distance than a parachuter if they jump from the same height .

Example: Draco.

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UNIT: 6

ANIMAL BEHAVIOUR

6.1. PARENTAL CARE –ARIUS



- Parental care can be defined as an association between the parents and the offsprings, so as to increase the chances of the survival of the young ones, and in fishes it includes all the post-spawning care of the offsprings by the parents.
- Most fishes do not care for their eggs or youngs and leave the spawning grounds soon after fertilisation.
- The lack of parental behaviour is correlated with production of great numbers of eggs and sperms. But there are many fishes where definite parental care has been evolved.
- Various devices have been adopted to ensure proper development of the eggs into adults. One or both the sexes may participate in the process.

- These include selection of a suitable site, nest building and various other methods of protection of the larvae.
- Species which do not exhibit any special device for safety of the ova, generally produce a very large number of eggs to increase the chances of survival of at least a few of them.
- Eggs of many species possess various mechanisms for attachment to stones, pebbles or aquatic vegetation, so that they are prevented from being washed away with the current of water.

Nest Building:

- Some fishes prepare crude nests for egg laying. At first a suitable place for preparing the nest is selected and some species may defend the place till death.
- Males of many species like the Darters (*Etheostoma*), sunfishes and cichlids, prepare a shallow basin- like nest for laying eggs by females.
- The stones and pebbles are removed from such nest and male keeps close watch over the eggs till hatching.
- A few species, however, leave the nest unprotected. Many freshwater fishes prepare crude nest with aquatic vegetation where eggs are laid.
- Protopterus and Lepidosiren prepare deep hole into which the females lay eggs. Males protect the nest till development is complete.

- **Amia calva (bowfin) prepare a crude circular nest among aquatic vegetation.**
- **The fertilised ova are protected by male who keeps guard over the nest till the young ones are hatched. The young ones are allowed to leave the nest in a body under the protection of father.**
- **Both the male and female of some cat fishes of North America prepare a crude nest in the mud for egg-laying. The nest is sometimes provided with protective cover of logs, stones, etc.**
- **Most interesting example is provided by the male stickleback *Gasterosteus aculeatus*, a small freshwater fish of North American lakes and ponds.**
- **The male fish actually builds a nest of dead aquatic weeds which are joined together by a sticky secretion produced from the kidneys.**
- **When the nest assumes a considerable size, the male makes a small tunnel. After the formation of tunnel and an elaborate courtship ritual, the male drags a mature female into the tunnel for laying eggs.**
- **After laying eggs, the female swims away and the male keeps watch over the fertilised eggs till development is over.**
- **In addition, foamy nest prepared by blowing of bubbles of air and sticky mucus are also encountered in many fishes. The bubbles of air and mucus adhere to form a floating mass of foam.**

- The eggs are collected by the male in his mouth cavity and he throws them in such a way that the eggs can adhere to the lower surface of foamy nest. This type of caring for eggs is found in Betta, Macropodus and many other fishes.

Mouth Cavity as Shelter:

- In some species, eggs develop within the mouth of the parent. In many cichlids, the female carries the eggs in her oral cavity.
- After hatching, the young fry do not leave the shelter for some time, and swim about in water very near the mouth, so that they can return to it in case of danger (found in Tilapia). In the cat fish, Arius the male carries the eggs and young ones in his mouth, and does not take food during this period.

Coiling Round Eggs:

- In butter fish, Pholis rolls the eggs into a rounded ball and then one of the parents remains on guard, possibly male, guards the egg-ball by coiling round it.

Attachment to Body:

- In Kurtus indicus (Perciformes) the male develops a bony hook projecting from the forehead and is supported by a special process of skull bone.
- The eggs are grouped in two bunches with the help of filamentous processes of the egg membrane.

- The eggs are attached to the hook of the forehead, in such a way that one bunch of eggs lies on either side of the head of the male, as he swims in water.

Formation of Integumentary Cups:

- In a cat fish, *Platystacus* of Brazil, shows an interesting method of parental case. During breeding season, the skin of lower surface of the body of the female fish becomes soft and spongy.
- Immediately after fertilisation of the eggs, the female presses her body against the eggs in such a way that each egg becomes lodged in a small integumentary depression.
- Each egg is attached inside the cup by an inconspicuous stalk. The eggs remain in this position till hatching.

Development of Brood Pouches:

- In the pipe fish, *Syngnathus* and the sea horse, *Hippocampus* the eggs develop within the brood pouch of the male.
- The eggs are transferred into the brood pouch by the female and development takes place within the brood pouch.

Mermaid's Purses:

- Oviparous sharks (e.g., *Scyllium*) lay fertilised eggs inside the protective horny egg capsules or mermaid's purses, which remain anchored to the sea

weeds by their long tendrils. The young hatch out after rupturing the egg case.

Viviparity:

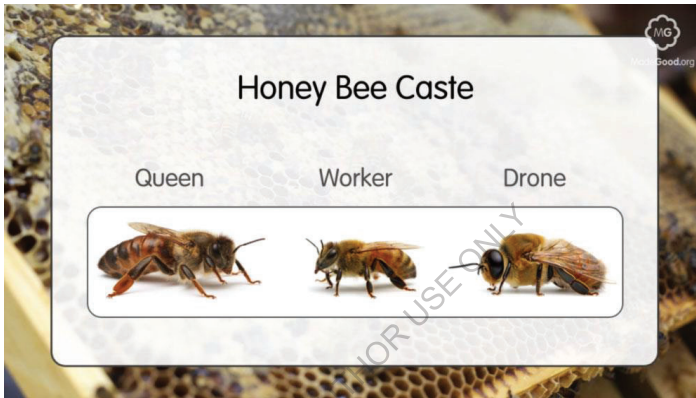
- The highest degree of parental care is found in viviparous fishes where young develop within the oviduct of the female.
- A few species are viviparous, such as the dog-fish, *Scoliodon* and the surf fish *Cymatogaster aggregatus*.
- Both fertilisation and development are internal. Developing embryos are nourished mostly by a yolk sac placenta and the young are born with the characteristic of the adult.
- Viviparity provides maximum protection and represents the highest degree of parental care.

UNIT: 7

ECONOMIC ZOOLOGY

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7.1. HONEY BEE CASTES



CASTES OF BEES:

In a hive or in the residence of bees these are three kinds of bees having different structure and function.

- a) Queen bee
- b) Male or king bee
- c) Worker bee

These three kinds of bees live together in the hive and do their own work independently:

a) Queen bee:

- **The queen is larger than the other bees. Laying eggs is the only function of the queen and she lays about 1500 eggs per day.**
- **Their life span is of three years. Each hive contains only one active queen bee.**
- **If any new queen is born in the hive, the old queen kills her by stinging.**

Male or king bees:

- **In a hive there may be a few male bees which take part in reproduction only. They are sting less and they do not collect honey.**

b) Worker bees:

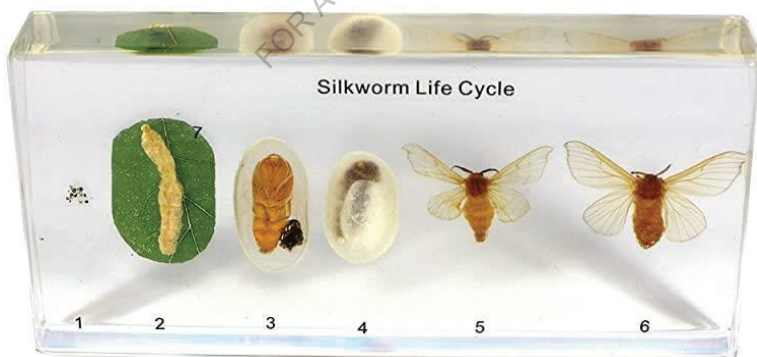
- **As a matter of fact, they are female bees but they are sterile and do not lay eggs.**
- **Their function is to collect honey, to look after the king, queen and larvae and to defend the hive from the enemies by stinging, to do overall supervision of the combs.**
- **They clean the chambers of the comb and produce new chambers by secreting wax from the body.**
- **Adult workers collect honey, pollen grains from the flowers to flowers and store in the comb.**

- The workers pollinate flowers indirectly when they roam about at the time of collecting honey and pollen grains.
- Due to pollination by bees, it is possible to produce many crops. and the production of many crops increases by 20-25%.

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7.2. SILKWORM-ADULT, CATERPILLAR, PUPA AND COCOON

- The life cycle of silk moth starts when a female silk moth lays eggs. The caterpillar or larvae are hatched from the eggs of the silk moth.
- The silkworms feed on mulberry leaves and give rise to pupa. In the pupa stage, a weave is netted around by the silkworm to hold itself.
- After that it swings its head, spinning a fibre made of a protein and becomes a silk fibre. Several caterpillars form a protective layer around pupa and this covering is known as the cocoon.
- The silk thread (yarn) is obtained from the silk moth's cocoon. The life cycle of the silkworm is explained below in detail.



Stage 1: Egg

- An egg is the first stage of the life cycle of the silkworm. The egg is laid by a female moth which is mostly the size of small dots.
- A female moth lays more than 350 eggs at a time. In the springtime, the eggs hatch due to the warmth in the air. This procedure happens once in every year.

Stage 2: Silkworm

- A hairy silkworm arises after the eggs crack. In this stage of silkworms, the growth happens.
- They feed on mulberry leaves and consume a large amount of these leaves for around 30 days before going to the next stage.

Stage 3: Cocoon

- In this stage, silkworms spin a protective cocoon around itself.
- It is the size of a small cotton ball and is made of a single thread of silk.

Stage 4: Pupa

- The pupa stage is a motionless stage.
- In this stage, people kill the pupa by plunging the cocoon into boiling water and unwind the silk thread.

Stage 5: Moth

- **In this stage, the pupa changes into an adult moth.**
- **The female moth lays eggs after mating and thus the life cycle of silkworm begins again.**

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7.3. PISCICULTURE –EDIBLE FISHES



- Fish farming or pisciculture involves commercial breeding of fish, usually for food, in fish tanks or artificial enclosures such as fish ponds.
- It is a particular type of aquaculture, which is the controlled cultivation and harvesting of aquatic animals such as fish, crustaceans, mollusca and so on, in natural or pseudo-natural environment.
- A facility that releases juvenile fish into the wild for recreational fishing or to supplement a species' natural numbers is generally referred to as a fish hatchery. Worldwide, the most important fish species produced in fish farming are carp, catfish, salmon and tilapia.
- Global demand is increasing for dietary fish protein, which has resulted in widespread overfishing in wild fisheries, resulting in significant decrease in fish stocks and even complete depletion in some regions.

- Fish farming allows establishment of artificial fish colonies that are provided with sufficient feeding, protection from natural predators and competitive threats, access to veterinarian service, and easier harvesting when needed, while being separate from and thus do not usually impact the sustainable yields of wild fish populations. While fish farming are practised worldwide, China alone provides 62% of the world's farmed fish production.
- As of 2016, more than 50% of seafood was produced by aquaculture. In the last three decades, aquaculture has been the main driver of the increase in fisheries and aquaculture production, with an average growth of 5.3 percent per year in the period 2000–2018, reaching a record 82.1 million tones in 2018.
- Farming carnivorous fish such as salmon, however, does not always reduce pressure on wild fisheries, as such farmed fish are usually fed fishmeal and fish oil extracted from wild forage fish.
- The 2008 global returns for fish farming recorded by the FAO totaled 33.8 million tonnes worth about US\$ 60 billion

UNIT: 8

BIO SAFETY IN LABORATORY

8.1. Bio safety in laboratory

1. No food or drink is allowed in any biology lab. No gum chewing. No putting on of makeup. Lunches, thermoses, and water bottles can be stored inside backpacks or purses.
2. They are not allowed on lab benches, the floor, near computers, or in sinks. Coffee cups are not allowed in the teaching and research labs. Be polite- no pushing, shoving, fighting or horseplay.
3. Always wear goggles when you work with chemicals, hot liquids, or other materials that could harm you. Use caution when working with organic solvents if you wear contact lenses (especially soft lenses) – if you're in doubt about what chemicals you're working with ask the instructor.
4. Wear gloves while working in the laboratory – gloves are not provided- *you must purchase examination gloves yourself*. Gloves are available for purchase at the campus bookstore. Please let us know if you have an allergy to latex.
5. If you have allergies or asthma or think you might be pregnant please inform the instructor.
6. Always wear closed toed shoes and long pants or a long skirt during labs to protect your legs and feet from spills.
7. Always alert your teacher to any spills on your skin or clothing.
8. Never taste or smell ANY chemicals unless your teacher tells you it is safe to do so.
9. Tie or pin long hair, scarves and headwear out of the way to avoid contact with flames or chemicals.

- 10. Be sure you know how to use equipment before beginning an experiment. If you are unsure - ASK !**
- 11. Children are not allowed in the laboratory. There are too many hidden dangers for young people.**
- 12. Mouth pipetting or drinking of solutions is strictly prohibited.**
- 13. Broken glassware or sharp metal pieces should be placed in “sharps” boxes (also labeled “broken glass”) unless it is contaminated with body fluids or microorganisms.**
- 12. Contaminated sharps should be placed in an appropriate labeled metal or glass container for sterilizing.**
- 14. Organic fluids (e.g., ether, acetone, chloroform) or other volatile liquids should be used inside a fume hood.**
- 15. Chemical wastes should never be placed in a sink drain without permission. Please consult your instructor in all cases.**
- 16. Media with microorganisms (liquid or solid) should be placed in a Biohazard bag or labeled autoclavable beakers for sterilizing and disposal.**
- 17. All glassware should be rinsed after use and test tubes placed upside down in the racks provided. Slides and coverslips should be discarded.**
- 18. Students and instructors should know the locations and operation of all safety Equipment. This includes fire extinguishers, eye washes, body showers, first aid kits, and the nearest exit from their room.**

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